UNIVERSITY OF LONDON

MSc Data Science

DSM010 Big Data Analysis

Final Coursework Assignment: Oct 2025 to Mar 2026 study session

- Please Note: You are permitted to upload your Coursework in the final submission area as many times as you like before the deadline. You will receive a similarity/originality score which represents what the Turnitin system identifies as work similar to another source. The originality score can take over 24 hours to generate, especially at busy times e.g. submission deadline.

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Coursework-2 Description

For this coursework, you will define your own topic based on your personal interests and background. Your topic must be suitable for a Big Data Analysis module and the scale of work should be appropriate for a distributed analytics project, meaning it should involve a dataset large enough to benefit from distributed processing, but manageable within the constraints of the Lena cluster. This coursework carries 70% of the total module mark.

While selecting the topic, please ensure that your project:

- **Core Requirements:** You must use Hadoop's HDFS for distributed data storage, Apache Spark/Spark MLlib for data analysis, and PySpark for coding. These form the backbone of the project.
- **Optional/Additional Technologies:** To reflect modern industry practices, you have the opportunity to integrate one or more of the following based on your project needs:
    - **MongoDB:** For projects that involve semi-structured or unstructured data (e.g., JSON documents, logs), you may use MongoDB as a supplement to HDFS.
    - **Kafka:** If your project involves real-time data ingestion or streaming, you can incorporate Apache Kafka into your data pipeline.
    - **PostgreSQL:** For handling structured data that requires robust, relational queries or transactional consistency, PostgreSQL is available as an option for storing and querying your processed results.

Note: The use of MongoDB, Kafka, and PostgreSQL is optional to successfully implement your chosen big data analysis project. You are free to choose any combination that best suits your dataset and analytical needs. In your proposal, clearly explain your rationale for using (or not using) each additional technology. You may still opt not to use one of MongoDB, Kafka, and PostgreSQL and complete the big data analysis project, but remember that using one of these technologies carries some marks.

There are **FOUR** main components for this coursework.

Component 1: Topic Proposal

- **Technologies and Integration Plan Section:** In addition to describing your dataset, hypotheses, and planned statistical analyses, include a section where you:
    - Describe your core tools (HDFS, Spark, and PySpark).
    - Discuss whether and why you might use MongoDB for managing any semi-structured data.
    - Explain if you plan to add Kafka for real-time data processing and outline what advantage it may bring.
    - Describe how PostgreSQL might be employed for structured data storage and ad-hoc queries.

Emphasise that you have the freedom to use these additional tools only if they add value to your chosen topic. Justify your decisions with clear reasoning.

Specifications: The topic proposal should be between 1,000 to 1,500 words, not including figures and tables. Your work should be fully referenced in a consistent reference style of your choosing. Any figures and tables should have captions and be referred to in the main text. A typical proposal might have the following sections: Introduction, Data (with summary stats), Hypotheses, Planned Analysis.

Component 2: Implementation of the Proposed Project

- **Flexible Integration Options:** Your implementation should demonstrate the core data analysis using PySpark, MLlib with HDFS storage. In addition, if you decide to work with:
    - **Kafka:** Present a code segment or module where you ingest simulated or live streaming data, process it via Spark, and then integrate it into your analysis.
    - **MongoDB:** Show how you might optionally store parts of the dataset that require a schema-less database approach, for example, saving intermediate JSON outputs.
    - **PostgreSQL:** Alternatively, use PostgreSQL if you wish to store processed, well-structured data to support complex queries.

Provide clear setup and configuration instructions for any additional technology you choose to include in your solution. Including your dataset with your submission is impossible. So, include the path to it on UoL's Lena cluster – either local files or HDFS location. Note that the zipped project you upload to Turnitin must stay under 10 MB; do *not* embed data files larger than that limit inside the archive. Or include a publicly accessible URL from which the examiners can get your dataset.

Your submission should contain instructions for the examiners to 1) understand what you did, and 2) run it. ***In situations where running your code on the full dataset takes a prohibitively long time, include instructions to run it on a subset of the data***. A reasonable subset is **≤ 10 MB of input data**, so the marker can reproduce results within the submission size cap.

Component 3: Limitations

- **Discussing the Trade-Offs:** In your 500-word discussion on limitations, include a section that evaluates:
    - The benefits of using additional tools like MongoDB, Kafka, or PostgreSQL.
    - Any challenges, such as increased configuration complexity or potential performance bottlenecks when integrating multiple systems.
    - Compare alternative scenarios. For example, if you had access to a high-performance computing cluster, which tool combination might be optimal?

This discussion should demonstrate that you understand when it's appropriate to use additional technologies and how they might scale with your project.

Component 4: Summary and Conclusions

- **Reflecting on Technology Choices:** In your final summary (500–1,000 words), add a brief discussion that reflects on:
    - What you explored in your project, the overall goals of your project, how you addressed your challenges, the results you obtained, and some discussion points.
    - The impact these tools had (if used) on your system's scalability, performance, and flexibility.

- Any lessons learned about choosing the right tool for the right task in a big data ecosystem.

This will show evaluators that you've critically thought about the choices and trade-offs in designing a modern distributed data analysis system.

Evaluation and Documentation

- **Guidance for Documentation:** Regardless of your integration choices, ensure that your README file and any accompanying documentation clearly state:
  - Which additional technologies you chose to use and why.
  - Detailed setup instructions, including configuration details (such as connection strings, topics, or schemas) for any optional tools.
  - How the different components (HDFS, Spark/PySpark, and any of MongoDB/Kafka/PostgreSQL) interact within your project.
  - Instructions on how to run your code—on the full dataset as well as on a subset if required.
  - Evidence of unaltered submission: provide either – an `ls -l` (Unix) or `Get-ChildItem -File` (PowerShell) directory listing of your project root, captured immediately before upload, or – a screenshot of the Lena cluster's file-browser showing the same timestamps. Repositories or hosts *without visible 'last-edited' timestamps* (e.g., some cloud notebooks) are **not acceptable**.
- **The submission will be assessed for**
  - Technical merit.
    Were the technical aspects employed in the project advanced and well-designed?
    Was the project technically challenging?

  - Analytic merit.
    Does the project answer complex, data-driven questions and / or provide interesting insights?

  - Documentation.

    Was the submission sufficiently documented?

The grades in this coursework are allocated approximately as follows:

Activity	Mark
• Aims and objectives of the topic for the big data analysis are clearly defined	6
• Storing the big data using HDFS	4
• Using Apache Spark's structured APIs	5
• Design and deploy machine learning pipelines with Apache Spark	7
• Technical challenge of the project	6

• Using MongoDB/Kafka/PostgreSQL	7
• Reflecting on technology choices	5
• Summarising/answering data driven questions related to the selected topic.	8
• Well-written PySpark code	8
• Limitations and Alternative solutions	6
• Overall quality of the report	8

Indicative Guidance:

- **Pass (50–59%)**: Basic implementation of HDFS and Spark; limited use of optional tools; minimal documentation.
- **Merit (60–69%)**: Clear project structure; appropriate use of Spark MLlib; some integration of optional tools; good documentation and analysis.
- **Distinction (70%+)**: Technically challenging project; innovative use of tools; critical analysis of design choices; excellent documentation and reproducibility.

For students aiming for higher marks, include a brief literature or industry review (with references) of similar big data analysis projects or pipelines. This should highlight how your approach compares or improves upon existing methods.

Sources for big data:

You may get access to big data sets from places like Kaggle https://www.kaggle.com/datasets or any other sources based on your choice of the topic.

Plagiarism:

Plagiarism is a form of cheating and is an academic offence. Do not be tempted and certainly do not succumb to temptation. Plagiarised copies are invariably rooted out and severe penalties apply. All assignment submissions are electronically tested for plagiarism.